



Design Patterns

SOLID · GoF (23) · DDD · Clean Architecture

Overview

10 sub-chapters covering all 23 GoF patterns plus SOLID, DDD, and Clean Architecture. Every pattern is implemented as a named class in ShopVerse.

SOLID Principles → ShopVerse Mapping

Principle	Rule	ShopVerse Example
S – Single Responsibility	One class, one job	OrderService: orchestration only; AuditService: audit writing only
O – Open/Closed	Open for extension, closed for modification	PricingStrategy interface → add FlashSalePricing without touching OrderService
L – Liskov Substitution	Subtype usable anywhere parent is expected	ExpandedCustomer extends Customer
I – Interface Segregation	Small, specific interfaces	Readable, Writable, Searchable separate; ProductRepository implements all
D – Dependency Inversion	Depend on abstractions	OrderService depends on PricingStrategy, not RegularPricing

Creational Patterns

Pattern	Class	Key Detail
Singleton	ApplicationConfig	Spring singleton scope
Factory Method	NotificationFactory.create(type)	Returns Email/Sms/PushNotification
Abstract Factory	DataAccessFactory	JpaOrderRepository vs InMemoryOrderRepository by profile
Builder	Order.Builder	OrderBuilder.forCustomer(id).withItem(sku,qty).build()
Prototype	OrderTemplate	clone() for repeat orders

Structural Patterns

Pattern	Class	Key Detail
Adapter	StripePaymentAdapter	Wraps Stripe SDK → PaymentGateway interface
Bridge	NotificationSender + NotificationChannel	Email/SmsSender are channels; Urgent/Routine are abstractions
Composite	CategoryNode	getTotalProductCount() recurses leaf+composite
Decorator	LoggingOrderService	Wraps OrderServiceImpl; logs entry/exit/duration
Facade	CheckoutFacade	checkout(cartId) calls Cart→Inventory→Payment→Order→Notification
Flyweight	CurrencySymbolCache	Shared immutable CurrencySymbol objects for USD/EUR/GBP
Proxy	LazyProductImageProxy	Loads full image URL only when getImageUrl() called (virtual proxy)

Behavioral Patterns

Pattern	Class	Key Detail
Chain of Responsibility	OrderValidationChain	Stock→Price→Fraud→AgeRestriction validators
Command	OrderCommandInvoker	PlaceOrderCommand, CancelOrderCommand, RefundOrderCommand
Iterator	ProductCatalogIterator	Iterates pages from DB without full load

Pattern	Class	Key Detail
Mediator	OrderEventBus	Services communicate through it, not directly
Memento	OrderStateMemento	Saves Order state before status change for rollback
Observer	OrderEventPublisher	InventoryListener, AuditListener, NotificationListener
State	OrderStateMachine	PENDING→CONFIRMED→PROCESSING→SHIPPED→DELIVERED; invalid trans
Strategy	PricingContext	RegularPricing, MemberPricing, FlashSalePricing by CustomerTier
Template Method	ReportGenerator	gatherData()→formatData()→writeOutput() abstract steps

DDD Building Blocks

DDD Concept	Implementation	Rule
Aggregate	Order (root); OrderItems added only via OrderItem	Don't let OrderItem manipulation from outside
Value Objects	Money(amount,currency), Address(...)	Immutable records; equals/hashCode by value
Domain Events	OrderPlacedEvent, OrderShippedEvent, OrderCancelledEvent	Can't delete aggregate, published by application service
Bounded Contexts	com.shopverse.ordering / catalog / inventory	Package names reflect ubiquitous language

Clean Architecture Layers

